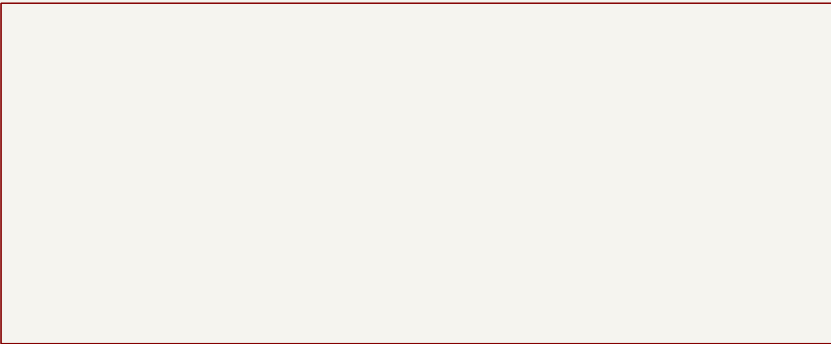


NESCART REV A – FUNCTIONAL SCHEMATIC

BUS CORE

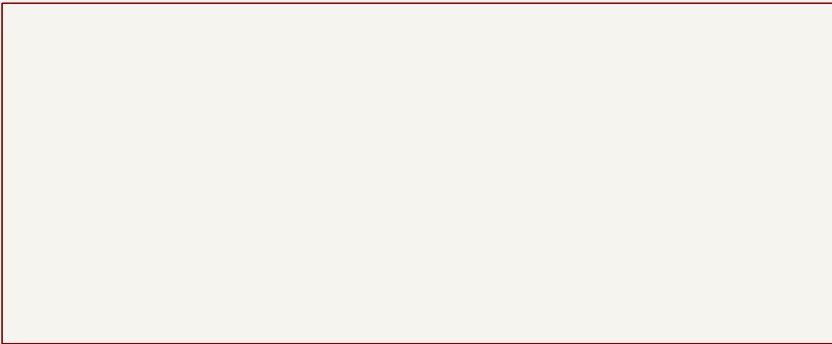
NES EDGE / MEMORY DATA PATH



ファイル: nescart_bus_core.kicad_sch

LOADER + INTERFACES

UPLOAD / ADDRESS GENERATION

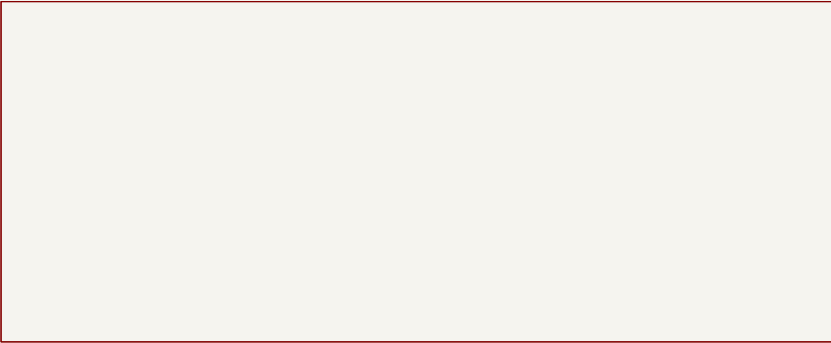


ファイル: nescart_loader.kicad_sch

BUS CORE <--> LOADER | all pages share named global nets

POWER + RESET

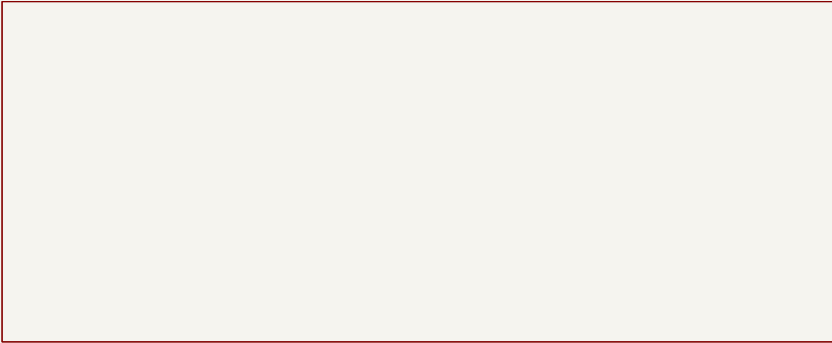
RESET / POWER / SAFE STATES



ファイル: nescart_power_reset.kicad_sch

DECOUPLING

LOCAL SUPPLY BYPASSING



ファイル: nescart_decoupling.kicad_sch

Functional flat hierarchy; global nets connect pages

kelta

Sheet: /

File: nescart.kicad_sch

Title: nescart Rev A – schematic index

Size: A4

Date:

Rev: A

KiCad E.D.A. 10.0.1

Id: 1/5

ESP32 LOADER / ADDRESS GENERATION / USB

ADDRESS SHIFT CHAIN

ESP32-S3

USB / DEBUG / USER I/O

FROM BUS CORE

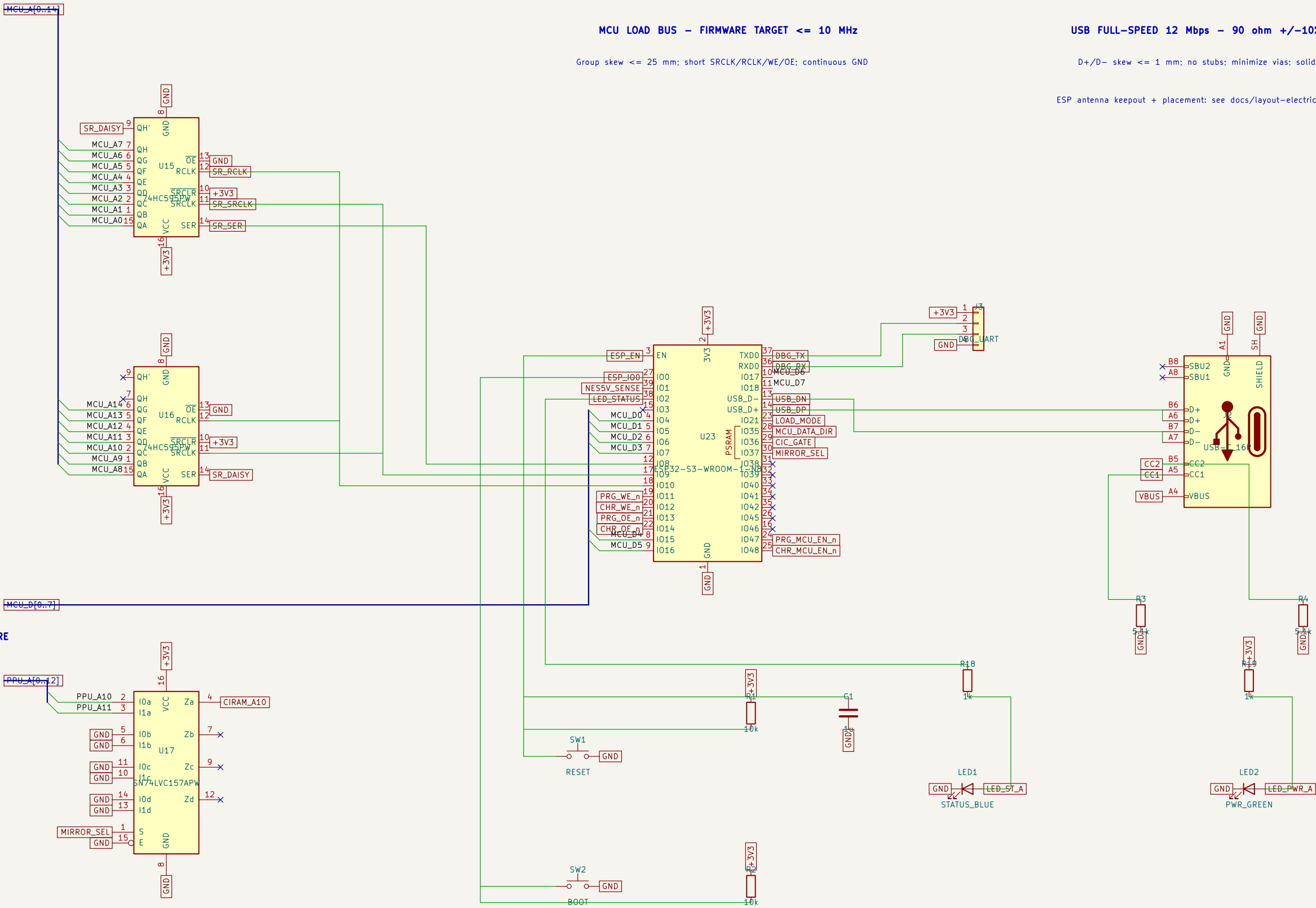
MCU LOAD BUS - FIRMWARE TARGET <= 10 MHz

Group skew <= 25 mm; short SRCLK/RCLK/WE/OE; continuous GND

USB FULL-SPEED 12 Mbps - 90 ohm +/-10% differential

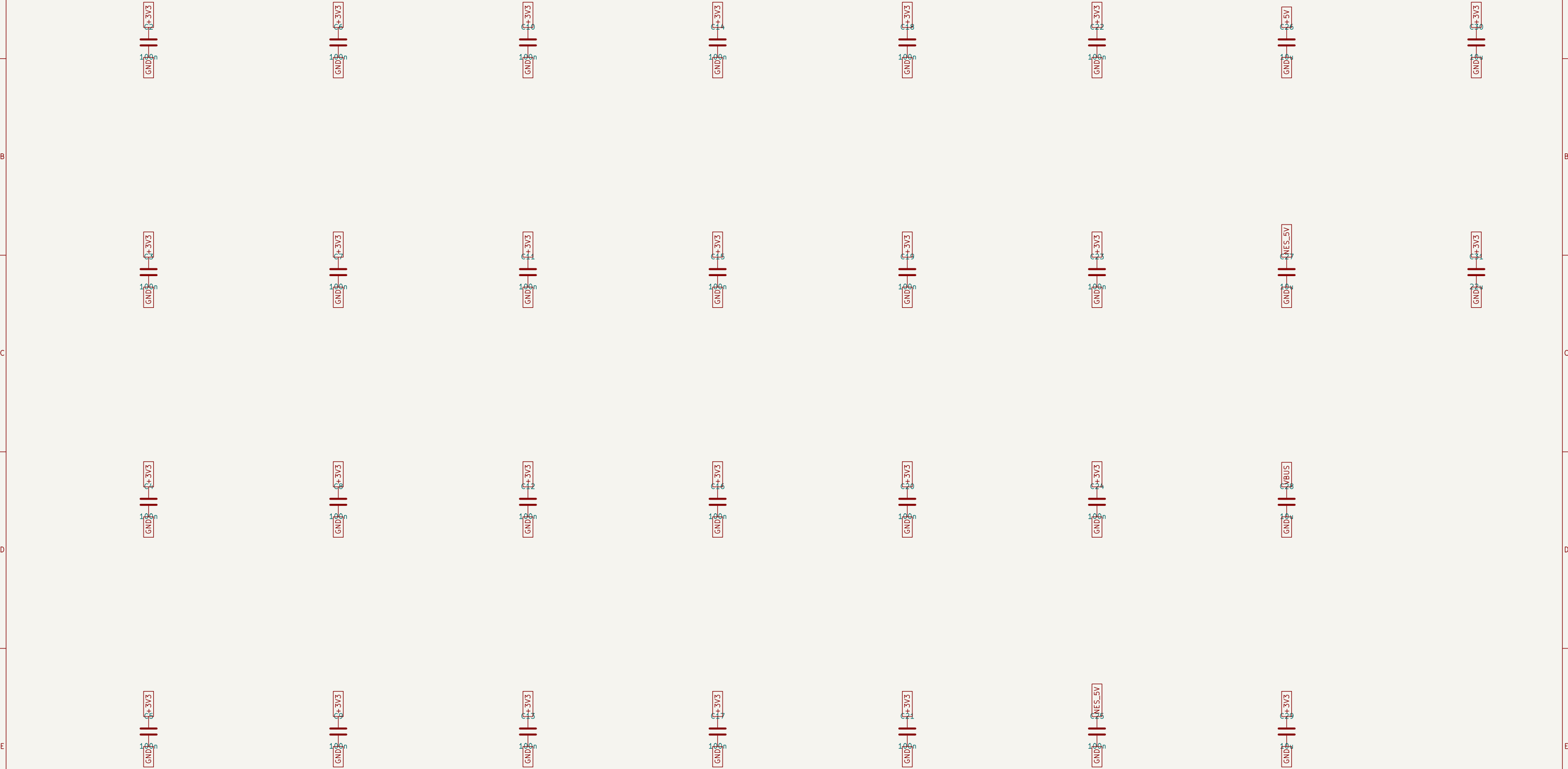
D+/D- skew <= 1 mm; no stubs; minimize vias; solid GND below

ESP antenna keepout + placement: see docs/layout-electrical-guidance.md



1	2	3	DECOUPLING / BULK CAPACITORS	6	7	8
---	---	---	------------------------------	---	---	---

One local capacitor per IC plus rail bulk capacitance



LAYOUT: each 100 nF in the VCC-pin escape path with its own short GND via

Place $\geq 10 \mu F$ at each source entry and directly at the ESP32 branch

Id: 5/5